

# Chapter 2 (Part 2): Auctions, Iterated Dominance, & Nash Equilibrium

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ECON 5001

Prof. Paul J. Healy

Based on: *Game Theory* by Giacomo Bonanno

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# Where We Left Off

We can now:

- separate a **frame** from a **game**
- write preferences as a complete, transitive  $\succsim_i$
- represent  $\succsim_i$  with an ordinal utility function
- compare two strategies of one player: **dominance**

And we found: if you have a dominant strategy, play it.

In this lecture: what if you don't?

## Log In: Auction Game

- MobLab: Two auctions. You're bidders.
- You'll be told what the item is worth **to you**
  - That number is yours alone; others have different ones
  - Random from \$10 to \$100 (equally likely)
- Groups of SIX bidders.
- Submit one sealed bid. Highest bid wins.
  - First game: Winner pays their bid
  - Second game: Winner pays highest *losing* bid (2nd highest bid)
- We'll play it **twice**, under two different rules

Your payoff is your value minus what you pay, or zero if you don't win.

No talking. Bid however you think is smart.

# What Did You Just Play?

**Round 1, first-price auction:** the winner pays *their own bid*.

**Round 2, second-price auction:** the winner pays *the 2nd highest bid*.

In which round did you bid closer to your true value?

Did you (bid < value) or (bid > value)?

For today we'll only focus on the **second-price** auction

## **Second-Price Auctions**

# The Rules

A **second-price** (or **Vickrey**) auction:

- sealed bids: nobody sees anyone else's bid
- highest bidder wins
- the winner pays the **second-highest** bid, not their own
- ties are broken by a rule fixed in advance

Why would a seller ever do this? It looks like giving money away.

## An Example: The Game-Day Permit

Suppose OSU auctions off a food truck permit near the stadium for game day. Dos Hermanos and Pitabilities both want it.

- Dos Hermanos (Player 1): estimates **\$30k** extra profit if they get it  
This is their **value** for the permit. Write  $v_1 = 30$  (in thousands)
- Suppose bids must be one of \$10k, \$20k, \$30k, \$40k, \$50k, \$60k  
So  $S_1 = S_2 = \{10, 20, 30, 40, 50, 60\}$ . They have 5 possible strategies.  
Write bids as  $b_1 \in S_1$  and  $b_2 \in S_2$
- **Second-price auction:** highest bid wins, pays *2nd highest bid*
- Ties go to Pitabilities (decided by a coin flip beforehand)

Dos Hermanos's payoff in the Second Price Auction:

$$\pi_1 = \begin{cases} v_1 - b_2 & \text{if } b_1 > b_2 \\ 0 & \text{if } b_1 \leq b_2 \end{cases}$$

Assume utility = money (selfish & greedy)

# Only Dos Hermanos's Payoffs Shown

|                             |    | Pitabilities |       |      |        |        |      |
|-----------------------------|----|--------------|-------|------|--------|--------|------|
|                             |    | 10           | 20    | 30   | 40     | 50     | 60   |
| Dos Hermanos ( $v_1 = 30$ ) | 10 | 0, ·         | 0, ·  | 0, · | 0, ·   | 0, ·   | 0, · |
|                             | 20 | 20, ·        | 0, ·  | 0, · | 0, ·   | 0, ·   | 0, · |
|                             | 30 | 20, ·        | 10, · | 0, · | 0, ·   | 0, ·   | 0, · |
|                             | 40 | 20, ·        | 10, · | 0, · | 0, ·   | 0, ·   | 0, · |
|                             | 50 | 20, ·        | 10, · | 0, · | -10, · | 0, ·   | 0, · |
|                             | 60 | 20, ·        | 10, · | 0, · | -10, · | -20, · | 0, · |

$$\text{Recall: } \pi_1 = \begin{cases} v_1 - b_2 & \text{if } b_1 > b_2 \\ 0 & \text{if } b_1 \leq b_2 \end{cases} \quad \text{with } v_1 = 30$$

## Reading the Table

- $b_1 = 30$  is a \_\_\_\_\_ strategy
- $b_1 = 40$  is a \_\_\_\_\_ strategy
- Strategies  $b_1 = 30$  and  $b_1 = 40$  are \_\_\_\_\_
- Overbidding ( $b_1 = 50$ ) is weakly dominated by  $b_1 = 30$ . Where is it strictly worse?
- Underbidding ( $b_1 = 20$ ) is weakly dominated by  $b_1 = 30$ . Where is it strictly worse?

# Vickrey's Theorem

**Theorem (Vickrey, 1961).** In a second-price auction, if player  $i$  is selfish and greedy, then bidding their true value ( $b_i = v_i$ ) is a **weakly dominant strategy**.

Remarkable, because:

- it holds no matter how many bidders there are
- it holds no matter what anyone else bids
- bidders don't need to guess, predict, or model their rivals at all!

**Lab result:** many people don't realize this, end up overbidding ( $b_i > v_i$ )  
“Failure of contingent thinking”

You played it... what do **you** think is going on??

## Why It Works: Two Cases

Let  $v_1$  be my value and  $b_2$  the other bid. Compare bidding  $v_1$  to anything else.

**Case 1:**  $b_2 \leq v_1$ . Bidding  $v_1$  wins and pays  $b_2$ , for a payoff of  $v_1 - b_2 \geq 0$ .

- Bidding higher: still win, still pay  $b_2$ . \_\_\_\_\_
- Bidding below  $b_2$ : lose, payoff 0. \_\_\_\_\_

**Case 2:**  $b_2 > v_1$ . Bidding  $v_1$  loses, for a payoff of 0.

- Bidding anything below  $b_2$ : still lose. \_\_\_\_\_
- Bidding at or above  $b_2$ : win, pay  $b_2$ . \_\_\_\_\_

Raising your bid can only ever change *whether* you win, never *what you pay*.

## Don't Forget: Preferences

The theorem assumes the bidder is **selfish and greedy**.

Drop that and it fails. Suppose I also care what *you* pay:

- **Spiteful:** if I lose, I'd rather you paid more
- **Benevolent:** if I lose, I'd rather you paid less

In either case, bidding my value is no longer dominant.

Again: *preferences matter!*

# **The Pivotal Mechanism**

# The Pivotal Mechanism

I'm skipping this due to time.

This will **not** appear on a quiz or exam.

In a nutshell:

- We're thinking about building a park.
- Person  $i$ 's value:  $v_i$ . Total cost:  $C$
- Social Welfare:  $W = \sum_i v_i - C$
- Pareto optimal: Build if  $W \geq 0$
- Suppose your tax is based on  $v_i$ . You'd lie about  $v_i$ !!

The fix: the **pivotal** mechanism

- Social Welfare if  $i$  didn't exist:  $W_{-i}$
- Player  $i$  is **pivotal** if  $W_{-i} < 0$  but  $W \geq 0$  (or vice-versa)
- Player  $i$ 's tax is  $W_{-i}$
- **Result:** It's a weakly dom. strategy to report  $v_i$  truthfully

Exactly like a 2nd-price auction!!



Vickrey, 1961



Clarke, 1971



Groves, 1973

- **1973:** Groves publishes the pivotal mechanism
- **1971:** Clarke had already published a special case
- **1961:** People realized it's just like Vickrey's 2nd price auction

The “pivotal mechanism” is often called “the **VCG** mechanism.”

Vickrey won the Nobel Prize in 1996 but died three days after it was announced.

**Using “Never Best Replies”  
to Find  
Strictly Dominated Strategies  
(not in the book)**

# Dominance vs. “Never Best Reply”

**Player 1:**  $M$  is strictly dominated by  $U$

How to find that? Try underlining best replies:

|          |   | Player 2 |      |      |
|----------|---|----------|------|------|
|          |   | L        | C    | R    |
| Player 1 | U | 4, 1     | 2, 0 | 4, 2 |
|          | M | 1, 2     | 1, 1 | 1, 0 |
|          | D | 0, 0     | 4, 2 | 0, 1 |

## Dominance vs. “Never Best Reply”

**Player 1:**  $M$  is strictly dominated by  $U$

How to find that? Try underlining best replies:

|          |   | Player 2     |                     |                     |
|----------|---|--------------|---------------------|---------------------|
|          |   | L            | C                   | R                   |
| Player 1 | U | <u>4</u> , 1 | 2, <u>0</u>         | <u>4</u> , <u>2</u> |
|          | M | 1, <u>2</u>  | 1, 1                | 1, <u>0</u>         |
|          | D | 0, <u>0</u>  | <u>4</u> , <u>2</u> | 0, 1                |

**Fact:** Strictly Dominated  $\Rightarrow$  Never a Best Reply to Any  $s_{-i}$  (never underlined)

Soooo... can we always use this to find strictly dominated strategies?

## Dominance vs. “Never Best Reply”

Use underlining to identify best replies to each column:

|          |   | Player 2            |                     |                     |
|----------|---|---------------------|---------------------|---------------------|
|          |   | L                   | C                   | R                   |
| Player 1 | U | 4, <u>1</u>         | <u>2</u> , <u>0</u> | 4, <u>2</u>         |
|          | M | 3, <u>2</u>         | 3, <u>1</u>         | 3, <u>0</u>         |
|          | D | <u>0</u> , <u>0</u> | 4, <u>2</u>         | <u>0</u> , <u>1</u> |

## Dominance vs. “Never Best Reply”

Use underlining to identify best replies to each column:

|          |   | Player 2     |                     |                     |
|----------|---|--------------|---------------------|---------------------|
|          |   | L            | C                   | R                   |
| Player 1 | U | <u>4</u> , 1 | 2, 0                | <u>4</u> , <u>2</u> |
|          | M | 3, <u>2</u>  | 3, 1                | 3, 0                |
|          | D | 0, 0         | <u>4</u> , <u>2</u> | 0, 1                |

M is **not** strictly dominated by U or D.

**Fact:** Never a Best Reply to any  $s_{-i} \not\Rightarrow$  Strictly Dominated

**Thus:** A “Never Best Reply” is a good *candidate* when finding strictly dominated strategies, but you still need to verify!

## An Example

Find the strictly dominated strategies for **Player 2**:

|          |   | Player 2 |      |      |      |      |      |
|----------|---|----------|------|------|------|------|------|
|          |   | A        | B    | C    | D    | E    | F    |
| Player 1 | W | 4, 5     | 1, 1 | 2, 3 | 0, 2 | 3, 0 | 2, 1 |
|          | X | 2, 1     | 2, 6 | 1, 3 | 4, 0 | 0, 4 | 3, 2 |
|          | Y | 1, 2     | 4, 1 | 3, 3 | 1, 4 | 2, 0 | 4, 0 |
|          | Z | 0, 0     | 0, 2 | 2, 3 | 3, 1 | 1, 1 | 1, 5 |

## An Example

Find the strictly dominated strategies for **Player 2**:

|          |   | Player 2            |              |              |              |              |              |
|----------|---|---------------------|--------------|--------------|--------------|--------------|--------------|
|          |   | A                   | B            | C            | D            | E            | F            |
| Player 1 | W | <u>4</u> , <u>5</u> | 1, 1         | 2, 3         | 0, 2         | <u>3</u> , 0 | 2, 1         |
|          | X | 2, <u>1</u>         | 2, <u>6</u>  | 1, 3         | <u>4</u> , 0 | 0, 4         | 3, 2         |
|          | Y | 1, 2                | <u>4</u> , 1 | <u>3</u> , 3 | 1, <u>4</u>  | 2, 0         | <u>4</u> , 0 |
|          | Z | 0, 0                | 0, 2         | 2, 3         | 3, 1         | 1, 1         | 1, <u>5</u>  |

## An Example

Find the strictly dominated strategies for **Player 2**:

|          |   | Player 2            |              |              |              |              |              |
|----------|---|---------------------|--------------|--------------|--------------|--------------|--------------|
|          |   | A                   | B            | C            | D            | E            | F            |
| Player 1 | W | <u>4</u> , <u>5</u> | 1, 1         | 2, 3         | 0, 2         | <u>3</u> , 0 | 2, 1         |
|          | X | 2, <u>1</u>         | 2, <u>6</u>  | 1, 3         | <u>4</u> , 0 | 0, 4         | 3, 2         |
|          | Y | 1, 2                | <u>4</u> , 1 | <u>3</u> , 3 | 1, <u>4</u>  | 2, 0         | <u>4</u> , 0 |
|          | Z | 0, 0                | 0, 2         | 2, 3         | 3, 1         | 1, 1         | 1, <u>5</u>  |

**Step 1:** Find all Never Best Replies

**Step 2:** Check which of those are strictly dominated.

Never Best Reply:  $\{C, E\}$

Strictly Dominated:  $\{E\}$

**Next Solution Concept:  
Iterated Deletion of Strictly Dominated Strategies**

# An Example

|          |   | Player 2 |      |      |
|----------|---|----------|------|------|
|          |   | L        | C    | R    |
| Player 1 | T | 4, 3     | 5, 1 | 6, 2 |
|          | M | 2, 1     | 8, 4 | 3, 6 |
|          | B | 3, 0     | 9, 6 | 2, 8 |

1. Underline best-response payoffs

# An Example

|          |   | Player 2            |              |              |
|----------|---|---------------------|--------------|--------------|
|          |   | L                   | C            | R            |
| Player 1 | T | <u>4</u> , <u>3</u> | 5, 1         | <u>6</u> , 2 |
|          | M | 2, 1                | 8, 4         | 3, <u>6</u>  |
|          | B | 3, 0                | <u>9</u> , 6 | 2, <u>8</u>  |

1. Underline best-response payoffs
2. There are no dominant strategies. But are any *strictly dominated*?
  - Fact: dominated  $\Rightarrow$  never a best response (not underlined)
  - So, check M and C
3. C is strictly dominated by \_\_\_\_\_. Delete it!!
  - Player 1 knows Player 2 will **not** play C

# An Example

|          |   | Player 2            |              |              |
|----------|---|---------------------|--------------|--------------|
|          |   | L                   | C            | R            |
| Player 1 | T | <u>4</u> , <u>3</u> | 5, 1         | <u>6</u> , 2 |
|          | M | 2, 1                | 8, 4         | 3, <u>6</u>  |
|          | B | 3, 0                | <u>9</u> , 6 | 2, <u>8</u>  |

1. Underline best-response payoffs
2. There are no dominant strategies. But are any *strictly dominated*?
  - Fact: dominated  $\Rightarrow$  never a best response (not underlined)
  - So, check M and C
3. C is strictly dominated by \_\_\_\_\_. Delete it!!
  - Player 1 knows Player 2 will **not** play C
4. But now M and B are strictly dominated by \_\_\_\_\_. Delete them

# An Example

|          |   | Player 2            |                             |              |
|----------|---|---------------------|-----------------------------|--------------|
|          |   | L                   | C                           | R            |
| Player 1 | T | <u>4</u> , <u>3</u> | 5, 1                        | <u>6</u> , 2 |
|          | M | 2, 1                | <del>8</del> , <del>4</del> | 3, <u>6</u>  |
|          | B | 3, 0                | <u>9</u> , <u>6</u>         | 2, <u>8</u>  |

1. Underline best-response payoffs
2. There are no dominant strategies. But are any *strictly dominated*?
  - Fact: dominated  $\Rightarrow$  never a best response (not underlined)
  - So, check M and C
3. C is strictly dominated by \_\_\_\_\_. Delete it!!
  - Player 1 knows Player 2 will **not** play C
4. But now M and B are strictly dominated by \_\_\_\_\_. Delete them
5. Now R is dominated by L. Delete R!

# An Example

|          |   | Player 2                           |                                           |                                    |
|----------|---|------------------------------------|-------------------------------------------|------------------------------------|
|          |   | L                                  | C                                         | R                                  |
| Player 1 | T | <u>4</u> , <u>3</u>                | 5, 1                                      | <u>6</u> , <u>2</u>                |
|          | M | 2, <u>1</u>                        | <del>8</del> , <del>4</del>               | <del>3</del> , <u><del>6</del></u> |
|          | B | <del>3</del> , <u><del>0</del></u> | <u><del>9</del></u> , <u><del>6</del></u> | <del>2</del> , <u><del>8</del></u> |

1. Underline best-response payoffs
2. There are no dominant strategies. But are any *strictly dominated*?
  - Fact: dominated  $\Rightarrow$  never a best response (not underlined)
  - So, check M and C
3. C is strictly dominated by \_\_\_\_\_. Delete it!!
  - Player 1 knows Player 2 will **not** play C
4. But now M and B are strictly dominated by \_\_\_\_\_. Delete them
5. Now R is dominated by L. Delete R!
6. Our solution:  $s^* = (T, L)$

Our new solution concept:

**I**terated  
**D**eletion of  
**S**trictly  
**D**ominated  
**S**trategies

(Also known as **IESDS**, for “Elimination” instead of “Deletion”)

## Iterated Deletion of Strictly Dominated Strategies.

Start with game  $G$ .

- $G^1$ : delete every strictly dominated strategy, for every player
- $G^2$ : do it again, inside  $G^1$
- $\vdots$
- $G^\infty$ : the output, reached in finitely many steps if  $G$  is finite

If  $G^\infty$  is a single strategy profile, we call it the **iterated strict dominant-strategy profile**. Otherwise it's just "the output of IDSDS."

We can justify IDSDS via “common knowledge of rationality”

1. I know you're rational  $\Rightarrow$  you won't play a strictly dominated strategy.  $G^1$
2. I know you know I'm rational  $\Rightarrow$  I know you'll eliminate my remaining strictly dominated strategies.  $G^2$
3. I know you know I know you're rational  $\Rightarrow$  I know you know that I'll eliminate your remaining strictly dominated strategies.  $G^3$
- ...
- $\infty$ . I know that you know that I know that...  $\Rightarrow$  IDSDS.  $G^\infty$

# Shortcoming of IDSDS

|          |   | Player 2 |      |      |      |
|----------|---|----------|------|------|------|
|          |   | W        | X    | Y    | Z    |
| Player 1 | A | 3, 4     | 6, 6 | 2, 1 | 2, 0 |
|          | B | 2, 1     | 1, 2 | 5, 5 | 2, 3 |
|          | C | 7, 3     | 3, 4 | 1, 3 | 2, 2 |
|          | D | 2, 8     | 1, 1 | 1, 2 | 9, 0 |

1. Underline best-response payoffs

# Shortcoming of IDSDS

|          |   | Player 2     |                     |                     |              |
|----------|---|--------------|---------------------|---------------------|--------------|
|          |   | W            | X                   | Y                   | Z            |
| Player 1 | A | 3, <u>4</u>  | <u>6</u> , <u>6</u> | 2, 1                | 2, 0         |
|          | B | 2, 1         | 1, 2                | <u>5</u> , <u>5</u> | 2, 3         |
|          | C | <u>7</u> , 3 | 3, <u>4</u>         | 1, 3                | 2, 2         |
|          | D | 2, <u>8</u>  | 1, 1                | 1, 2                | <u>9</u> , 0 |

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
  - Dominated  $\Rightarrow$  never a best response. Only Z has no underline
3. Z is strictly dominated by \_\_\_\_\_. Delete it!

# Shortcoming of IDSDS

|          |   | Player 2     |                     |                     |              |
|----------|---|--------------|---------------------|---------------------|--------------|
|          |   | W            | X                   | Y                   | Z            |
| Player 1 | A | 3, 4         | <u>6</u> , <u>6</u> | 2, 1                | 2, <u>0</u>  |
|          | B | 2, <u>1</u>  | 1, 2                | <u>5</u> , <u>5</u> | 2, 3         |
|          | C | <u>7</u> , 3 | 3, <u>4</u>         | 1, 3                | 2, 2         |
|          | D | 2, <u>8</u>  | 1, 1                | 1, 2                | <u>9</u> , 0 |

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
  - Dominated  $\Rightarrow$  never a best response. Only Z has no underline
3. Z is strictly dominated by \_\_\_\_\_. Delete it!
4. D's only underline was in Z. Now it is dominated by \_\_\_\_\_. Delete it!

# Shortcoming of IDSDS

|          |              | Player 2               |                     |                     |                        |
|----------|--------------|------------------------|---------------------|---------------------|------------------------|
|          |              | W                      | X                   | Y                   | Z                      |
| Player 1 | A            | 3, 4                   | <u>6</u> , <u>6</u> | 2, 1                | 2, 0                   |
|          | B            | 2, 1                   | 1, 2                | <u>5</u> , <u>5</u> | 2, 3                   |
|          | C            | <u>7</u> , 3           | 3, <u>4</u>         | 1, 3                | 2, 2                   |
|          | <del>D</del> | <del>2, <u>8</u></del> | <del>1, 1</del>     | <del>1, 2</del>     | <del><u>9</u>, 0</del> |

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
  - Dominated  $\Rightarrow$  never a best response. Only Z has no underline
3. Z is strictly dominated by \_\_\_\_\_. Delete it!
4. D's only underline was in Z. Now it is dominated by \_\_\_\_\_. Delete it!
5. W's only underline was in D. Now it is dominated by \_\_\_\_\_. Delete it!

# Shortcoming of IDSDS

|          |              | Player 2               |                     |                     |                        |
|----------|--------------|------------------------|---------------------|---------------------|------------------------|
|          |              | W                      | X                   | Y                   | Z                      |
| Player 1 | A            | 3, <u>4</u>            | <u>6</u> , <u>6</u> | 2, 1                | 2, 0                   |
|          | B            | 2, 1                   | 1, 2                | <u>5</u> , <u>5</u> | 2, 3                   |
|          | C            | <u>7</u> , 3           | 3, <u>4</u>         | 1, 3                | 2, 2                   |
|          | <del>D</del> | <del>2, <u>8</u></del> | <del>1, 1</del>     | <del>1, 2</del>     | <del><u>9</u>, 0</del> |

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
  - Dominated  $\Rightarrow$  never a best response. Only Z has no underline
3. Z is strictly dominated by \_\_\_\_\_. Delete it!
4. D's only underline was in Z. Now it is dominated by \_\_\_\_\_. Delete it!
5. W's only underline was in D. Now it is dominated by \_\_\_\_\_. Delete it!
6. C's only underline was in W. Now it is dominated by \_\_\_\_\_. Delete it!

# Shortcoming of IDSDS

|          |   | Player 2            |                     |                     |              |
|----------|---|---------------------|---------------------|---------------------|--------------|
|          |   | W                   | X                   | Y                   | Z            |
| Player 1 | A | 3, <u>4</u>         | <u>6</u> , <u>6</u> | 2, 1                | 2, 0         |
|          | B | 2, <u>1</u>         | 1, 2                | <u>5</u> , <u>5</u> | 2, 3         |
|          | C | <u>7</u> , <u>3</u> | 3, <u>4</u>         | 1, 3                | 2, 2         |
|          | D | 2, <u>8</u>         | 1, 1                | 1, 2                | <u>9</u> , 0 |

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
  - Dominated  $\Rightarrow$  never a best response. Only Z has no underline
3. Z is strictly dominated by \_\_\_\_\_. Delete it!
4. D's only underline was in Z. Now it is dominated by \_\_\_\_\_. Delete it!
5. W's only underline was in D. Now it is dominated by \_\_\_\_\_. Delete it!
6. C's only underline was in W. Now it is dominated by \_\_\_\_\_. Delete it!
7. Nothing left to delete:  $G^\infty$  is a  $2 \times 2$  game. IDSDS is not **decisive**.

# MobLab: Keynesian Beauty Contest

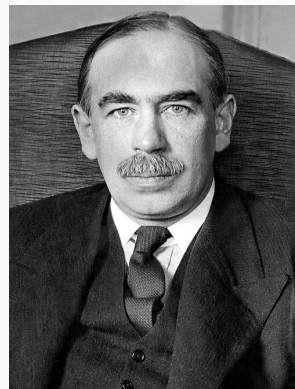
MobLab: “Beauty contest”  
Choose an integer from **1** to **100**.

Whoever is closest to  
 $\frac{2}{3} \times$  (the average of everyone’s number)  
wins.

**Origin:** Newspaper beauty contest games

**Keynes:** Similar to the stock market

**This Version:** match  $\frac{2}{3}$  the average, not the average



John Maynard Keynes

# Solving this Game



$\frac{2}{3}$  of the average can never exceed  $\frac{2}{3} \times 100 \approx 66.7$ .

- So 67 strictly dominates everything above it. Delete 68–100.
- Now the target can't exceed  $\frac{2}{3} \times 67 \approx 44.7$ . Delete 45–67.
- Now it can't exceed  $\frac{2}{3} \times 45 = 30$ . Delete 31–45.
- Keep going ... where does it end?

**Behavioral Game Theory:** Why does IDSDS fail in reality?

**Next Solution Concept: IDWDS**  
**...and Why It's Not Great**

# IDWDS Problem 1: Not Uniquely Defined

What if we delete **weakly** dominated strategies too?

|          |   | Player 2 |      |
|----------|---|----------|------|
|          |   | L        | R    |
| Player 1 | A | 4, 0     | 0, 0 |
|          | T | 3, 2     | 2, 2 |
|          | M | 1, 1     | 0, 0 |
|          | B | 0, 0     | 1, 1 |

Two perfectly reasonable places to start:

- $M$  is strictly dominated by  $T$  ← delete this first
- $B$  is strictly dominated by  $T$

# IDWDS Problem 1: Not Uniquely Defined

What if we delete **weakly** dominated strategies too?

|          |   | Player 2 |      |
|----------|---|----------|------|
|          |   | L        | R    |
| Player 1 | A | 4, 0     | 0, 0 |
|          | T | 3, 2     | 2, 2 |
|          | M | 1, 1     | 0, 0 |
|          | B | 0, 0     | 1, 1 |

Two perfectly reasonable places to start:

- $M$  is strictly dominated by  $T$
- $B$  is strictly dominated by  $T$  ← delete this first

## IDWDS Problem 2: Might be a Best Reply

Does IDWDS have the same rationality foundation?

|   | L     | R       |
|---|-------|---------|
| U | 50, 0 | 10, 100 |
| D | 10, 0 | 10, 100 |

$D$  is weakly dominated but is it *definitely* irrational?

Golden Balls example:

|       |   | Steven |        |
|-------|---|--------|--------|
|       |   | C      | D      |
| Sarah | C | 50, 50 | 0, 100 |
|       | D | 100, 0 | 0, 0   |

**Our Main Solution Concept:  
Nash Equilibrium**

## Still Not Enough

We need a solution concept that captures “rationality” but doesn’t require dominance.

**Dominance:**  $s_i^*$  is a best reply *no matter what others play*

**Nash Equilibrium:**  $s_i^*$  is a best reply *to what the others are actually playing*

- If  $(s_1^*, s_2^*)$  is a Nash equilibrium then:
  1.  $s_1^*$  is a best reply to  $s_2^*$ , and
  2.  $s_2^*$  is a best reply to  $s_1^*$

In Nash equilibrium, players **know** (or, correctly predict) what the others will do

Thus, a Nash equilibrium is **stable**. Nobody will deviate (*on their own*)

## An Example

|    |   | P2   |      |
|----|---|------|------|
|    |   | L    | R    |
| P1 | U | 1, 1 | 0, 0 |
|    | D | 0, 0 | 2, 2 |

Profile  $(U, L)$  is **stable**:

1. Player 1 is hurt by deviating from  $U$  to  $D$
2. Player 2 is hurt by deviating from  $L$  to  $R$
3. Note: *joint* deviation to  $(D, R)$  isn't considered. Only single-player deviations.

## An Example

|    |   | P2   |      |
|----|---|------|------|
|    |   | L    | R    |
| P1 | U | 1, 1 | 0, 0 |
|    | D | 0, 0 | 2, 2 |

Profile  $(U, L)$  is **stable**:

1. Player 1 is hurt by deviating from  $U$  to  $D$
2. Player 2 is hurt by deviating from  $L$  to  $R$
3. Note: *joint* deviation to  $(D, R)$  isn't considered. Only single-player deviations.

Profile  $(D, R)$  is also **stable**:

1. Player 1 is hurt by deviating from  $D$  to  $U$
2. Player 2 is hurt by deviating from  $R$  to  $L$

## Definition: Nash Equilibrium (Two Players)

In a 2-player game, a strategy profile  $s^* = (s_1^*, s_2^*)$  is a **Nash equilibrium** if

1. for every  $s_1 \in S_1$ :  $\pi_1(s_1^*, s_2^*) \geq \pi_1(s_1, s_2^*)$
2. for every  $s_2 \in S_2$ :  $\pi_2(s_1^*, s_2^*) \geq \pi_2(s_1^*, s_2)$

In words: **assuming the other player sticks to the Nash equilibrium**, neither player can do better by changing their strategy.

Does not allow **joint deviations** (see previous example)

## John Forbes Nash Jr. (1928–2015)



- Grew up in Bluefield, WV
- Undergrad: Carnegie Mellon
- Duffin's letter, 1948:  
"He is a mathematical genius."
- Princeton PhD, 1950: 26 pages
  - His equilibrium concept
- Struggled with delusions, schizophrenia
- Nobel Prize in Economics, 1994
- Subject of *A Beautiful Mind*, 2001
- I hung out with him for a few days, 2007
- Abel Prize (math), 2015, for PDEs
- Died in a taxi crash, May 2015
  - Was coming home from the Abel ceremony

## Another Example

|          |   | Player 2 |      |      |
|----------|---|----------|------|------|
|          |   | L        | C    | R    |
| Player 1 | T | 3, 2     | 0, 0 | 1, 1 |
|          | M | 3, 0     | 1, 5 | 4, 4 |
|          | B | 1, 0     | 2, 3 | 3, 0 |

Verify  $(T, L)$  is a Nash equilibrium (NE):

- Player 1, holding  $L$  fixed:  $T$  gives 3;  $M$  gives 3;  $B$  gives 1 ✓
  - Nothing is **strictly** better. (Ties are okay.)
- Player 2, holding  $T$  fixed:  $L$  gives 2;  $C$  gives 0;  $R$  gives 1 ✓

So  $(T, L)$  is a Nash equilibrium. Find the other one.

## Four Justifications

**No regret.** After seeing what the other player did, nobody wishes they had chosen differently.

**Self-enforcing agreement.** If the players could talk beforehand and shake on  $s^*$ , nobody would want to break it.

**Viable recommendation.** If a third party publicly recommended  $s^*$ , nobody would want to deviate, assuming the others comply.

**Rationality with correct beliefs.** Each player has a correct prediction to what the opponents will do, and best responds to that.

# Best Replies

Equivalent definition of Nash equilibrium:

$(s_1^*, s_2^*)$  is a Nash equilibrium  $\iff$  every  $s_i^*$  is a best reply to  $s_{-i}^*$

In math notation:

Strategy  $s_1$  is a **best reply** to  $s_2$  if

$$\pi_1(s_1, s_2) \geq \pi_1(s'_1, s_2) \quad \forall s'_1 \in S_1$$

Define  $BR_1(s_2) = \{s_1 : \pi_1(s_1, s_2) \geq \pi_1(s'_1, s_2) \quad \forall s'_1 \in S_1\}$ : PLR1's best replies to  $s_2$

Define  $BR_2(s_1) = \{s_2 : \pi_2(s_1, s_2) \geq \pi_2(s_1, s'_2) \quad \forall s'_2 \in S_2\}$ : PLR2's best replies to  $s_1$

Nash equilibrium:

$$(s_1^*, s_2^*) \text{ is a N.E. } \iff s_1^* \in BR_1(s_2^*) \text{ and } s_2^* \in BR_2(s_1^*)$$

Lets us use the underlining trick to find N.E.

# The Underlining Trick

1. In each **column**, underline Player 1's largest payoff (incl. ties)
2. In each **row**, underline Player 2's largest payoff (incl. ties)
3. Any cell with **both** numbers underlined is a Nash equilibrium

|          |   | Player 2 |      |      |      |
|----------|---|----------|------|------|------|
|          |   | E        | F    | G    | H    |
| Player 1 | A | 4, 0     | 3, 2 | 2, 3 | 4, 1 |
|          | B | 4, 2     | 2, 1 | 1, 2 | 0, 2 |
|          | C | 3, 6     | 5, 5 | 3, 1 | 5, 0 |
|          | D | 2, 3     | 3, 2 | 1, 2 | 3, 3 |

Player 1:  $BR_1(E) =$                        $BR_1(F) =$                        $BR_1(G) =$                        $BR_1(H) =$

Player 2:  $BR_2(A) =$                        $BR_2(B) =$                        $BR_2(C) =$                        $BR_2(D) =$

# The Underlining Trick

|          |   | Player 2     |              |              |              |
|----------|---|--------------|--------------|--------------|--------------|
|          |   | E            | F            | G            | H            |
| Player 1 | A | <u>4</u> , 0 | 3, 2         | 2, <u>3</u>  | 4, 1         |
|          | B | 4, <u>2</u>  | 2, 1         | 1, <u>2</u>  | 0, <u>2</u>  |
|          | C | 3, <u>6</u>  | <u>5</u> , 5 | <u>3</u> , 1 | <u>5</u> , 0 |
|          | D | 2, <u>3</u>  | 3, 2         | 1, 2         | 3, <u>3</u>  |

Only (B, E) has both payoffs underlined: the unique Nash equilibrium profile

$$B \in BR_1(E) \quad E \in BR_2(B)$$

Any other profile: someone would *gain (strictly)* by deviating *unilaterally*.

**Ex:** from (A, E), PLR2 would deviate to G

## More Than Two Players

A profile  $s^* = (s_1^*, s_2^*, \dots, s_n^*) \in S$  is a Nash equilibrium if, for every player  $i$ ,

$$\pi_i(s^*) \geq \pi_i(s_1^*, \dots, s_{i-1}^*, s_i, s_{i+1}^*, \dots, s_n^*) \quad \text{for all } s_i \in S_i$$

Equivalently (and more compact):

$$s_i^* \in BR_i(s_{-i}^*) \quad \text{for all } i \in \{1, 2, \dots, n\}$$

## A Three-Player Example

Let's try a 3-player game ( $2 \times 2 \times 2$ )

|          |   | Player 2 |         |
|----------|---|----------|---------|
|          |   | L        | R       |
| Player 1 | T | 4, 1, 4  | 0, 0, 1 |
|          | B | 3, 1, 3  | 3, 3, 2 |

Player 3: *E*

|          |   | Player 2 |         |
|----------|---|----------|---------|
|          |   | L        | R       |
| Player 1 | T | 0, 0, 1  | 2, 2, 3 |
|          | B | 1, 4, 2  | 1, 0, 3 |

Player 3: *F*

Two Nash equilibria:

## A Three-Player Example

Let's try a 3-player game ( $2 \times 2 \times 2$ )

|          |   | Player 2                       |                         |
|----------|---|--------------------------------|-------------------------|
|          |   | L                              | R                       |
| Player 1 | T | <u>4</u> , <u>1</u> , <u>4</u> | 0, 0, 1                 |
|          | B | 3, 1, <u>3</u>                 | <u>3</u> , <u>3</u> , 2 |

Player 3: *E*

|          |   | Player 2                |                                |
|----------|---|-------------------------|--------------------------------|
|          |   | L                       | R                              |
| Player 1 | T | 0, 0, 1                 | <u>2</u> , <u>2</u> , <u>3</u> |
|          | B | <u>1</u> , <u>4</u> , 2 | 1, 0, <u>3</u>                 |

Player 3: *F*

Two Nash equilibria:

## A Three-Player Example

Let's try a 3-player game ( $2 \times 2 \times 2$ )

|          |   | Player 2                       |                         |
|----------|---|--------------------------------|-------------------------|
|          |   | L                              | R                       |
| Player 1 | T | <u>4</u> , <u>1</u> , <u>4</u> | 0, 0, 1                 |
|          | B | 3, 1, <u>3</u>                 | <u>3</u> , <u>3</u> , 2 |
|          |   | Player 3: <i>E</i>             |                         |

|          |   | Player 2                |                                |
|----------|---|-------------------------|--------------------------------|
|          |   | L                       | R                              |
| Player 1 | T | 0, 0, 1                 | <u>2</u> , <u>2</u> , <u>3</u> |
|          | B | <u>1</u> , <u>4</u> , 2 | 1, 0, <u>3</u>                 |
|          |   | Player 3: <i>F</i>      |                                |

Two Nash equilibria:

$(T, L, E)$  and  $(T, R, F)$ : the only cells with all three payoffs underlined.

Check: Why isn't  $(B, R, E)$  and Nash equilibrium? How about  $(B, L, F)$ ?

What's the plural of “equilibrium”???

1. **Equilibria.** True to Latin pluralization
2. **Equilibriums.** Acceptable in modern English dictionaries.

**This class:** Equilibria.

# Non-Existence of Nash Equilibrium

Game: “Matching Pennies”

|          |   | Attacker     |              |
|----------|---|--------------|--------------|
|          |   | H            | T            |
| Defender | H | <u>1</u> , 0 | 0, <u>1</u>  |
|          | T | 0, <u>1</u>  | <u>1</u> , 0 |

Classic example of an “attacker-defender” (discoordination) game.

**Examples:** Penalty kicks, play-calling, tax auditing, law/rule enforcement

No cell has both payoffs underlined  $\Rightarrow \nexists$  Nash equilibrium!

With only ordinal payoffs, a game **may not have** a Nash equilibrium.

**Chapter 5:** if we allow players to randomize 50/50, **that** becomes a Nash equilibrium.

**Chapter 2:** no randomization yet. Only “pure strategies.”

**A Theorem:**

**Nash Equilibrium is never eliminated by IDSDS**

# Nash Survives IDSDS

Solution concepts predict a **set** of strategy profiles for a given game.

Math notation:

- $SDS(G)$ : the set of strictly dominant strategy profiles in game  $G$
- $IDSDS(G)$ : the set of profiles that survive IDSDS in game  $G$
- $IDWDS(G)$ : the set of profiles that survive IDWDS in game  $G$  (which order?)
- $NE(G)$ : the set of Nash equilibrium profiles for game  $G$

Each is a subset of  $S$ . Can be empty, can be a singleton, can have many profiles.

**Theorem.** For every ordinal strategic-form game  $G$ ,

$$NE(G) \subseteq IDSDS(G)$$

- A Nash equilibrium never uses a strategy that IDSDS deletes
- So IDSDS never throws away a Nash equilibrium
- But the same is **not** true of IDWDS: it is possible that  $NE(G) \cap IDWDS(G) = \emptyset$

## **Another Solution Concept: Strict Nash Equilibrium**

**(Technically, a “refinement” of N.E.)**

# Strict Nash Equilibrium

$s^*$  is a **strict** Nash equilibrium if every player is *strictly* worse off.

In other words, Nash equilibrium with **no ties allowed**.

$s^*$  is **Nash Eq.:**  $\pi_i(s_i^*, s_{-i}^*) \geq \pi_i(s_i, s_{-i}^*)$  for every  $s_i \in S_i$  (including  $s_i = s_i^*$ )

$s^*$  is **Strict Nash Eq.:**  $\pi_i(s_i^*, s_{-i}^*) > \pi_i(s_i, s_{-i}^*)$  for every  $s_i \neq s_i^*$

Two changes:  $\geq$  becomes  $>$ , and  $s_i^*$  has to be excluded on RHS

|   | L    | C    | R    |
|---|------|------|------|
| U | 3, 3 | 3, 1 | 0, 0 |
| M | 1, 0 | 1, 3 | 2, 1 |
| D | 0, 1 | 2, 0 | 2, 3 |

**Theorem:** All Strict Nash equilibria are Nash equilibria (  $SNE(G) \subseteq NE(G)$  )

# The IDSDS vs. Nash Theorem

We learned that  $NE(G) \subseteq IDSDS(G)$  for Nash equilibrium.

**Theorem:**  $SNE(G) \subseteq NE(G) \subseteq IDSDS(G)$ .

When are they all equal?

**Theorem.** If  $IDSDS(G) = \{s^*\}$  then  $s^*$  is a Strict Nash equilibrium and it is the only Nash equilibrium.

In math notation:

$$IDSDS(G) = \{s^*\} \implies SNE(G) = NE(G) = IDSDS(G) = \{s^*\}$$

**Putting It Together:  
IDSDS, IDWDS, NE, and SNE**

## What About IDWDS vs. NE and SNE?

How does IDWDS relate to Nash equilibrium and Strict Nash equilibrium?

|    |   | P2   |      |
|----|---|------|------|
|    |   | L    | R    |
| P1 | U | 1, 1 | 0, 0 |
|    | D | 0, 0 | 0, 0 |

$IDSDS(G) =$

$IDWDS(G) =$

$NE(G) =$

$SNE(G) =$

**Theorem:**

But what about  $NE(G)$  vs.  $IDWDS(G)$ ??

## What About IDWDS vs. NE and SNE?

How does IDWDS relate to Nash equilibrium and Strict Nash equilibrium?

|    |   | P2   |      |
|----|---|------|------|
|    |   | L    | R    |
| P1 | U | 1, 1 | 0, 0 |
|    | D | 0, 0 | 0, 0 |

$$IDSDS(G) = S$$

$$IDWDS(G) =$$

$$NE(G) =$$

$$SNE(G) =$$

**Theorem:**

But what about  $NE(G)$  vs.  $IDWDS(G)$ ??

## What About IDWDS vs. NE and SNE?

How does IDWDS relate to Nash equilibrium and Strict Nash equilibrium?

|    |   | P2   |      |
|----|---|------|------|
|    |   | L    | R    |
| P1 | U | 1, 1 | 0, 0 |
|    | D | 0, 0 | 0, 0 |

$$IDSDS(G) = S$$

$$IDWDS(G) = \{(U, L)\}$$

$$NE(G) =$$

$$SNE(G) =$$

**Theorem:**

But what about  $NE(G)$  vs.  $IDWDS(G)$ ??

## What About IDWDS vs. NE and SNE?

How does IDWDS relate to Nash equilibrium and Strict Nash equilibrium?

|    |   | P2   |      |
|----|---|------|------|
|    |   | L    | R    |
| P1 | U | 1, 1 | 0, 0 |
|    | D | 0, 0 | 0, 0 |

$$IDSDS(G) = S \quad IDWDS(G) = \{(U, L)\} \quad NE(G) = \{(U, L), (D, R)\} \quad SNE(G) =$$

**Theorem:**

But what about  $NE(G)$  vs.  $IDWDS(G)$ ??

## What About IDWDS vs. NE and SNE?

How does IDWDS relate to Nash equilibrium and Strict Nash equilibrium?

|    |   | P2   |      |
|----|---|------|------|
|    |   | L    | R    |
| P1 | U | 1, 1 | 0, 0 |
|    | D | 0, 0 | 0, 0 |

$$IDSDS(G) = S \quad IDWDS(G) = \{(U, L)\} \quad NE(G) = \{(U, L), (D, R)\} \quad SNE(G) = \{(U, L)\}$$

**Theorem:**  $SNE(G) \subseteq IDWDS(G)$  (Strict Nash always survives IDWDS)

But what about  $NE(G)$  vs.  $IDWDS(G)$ ??

## What About IDWDS vs. NE and SNE?

Now try a modified matching pennies (attacker-defender) game:

|   | H    | T      | D     |
|---|------|--------|-------|
| H | 1, 0 | 0, 1   | 0, -1 |
| T | 0, 1 | 1, 0   | 0, 0  |
| D | 0, 0 | -1, -1 | 0, 0  |

$$IDSDS(G) =$$

$$NE(G) =$$

$$IDWDS(G) =$$

$$SNE(G) =$$

**Theorem:**  $SNE(G) \subseteq IDWDS(G)$  (Strict Nash always survives IDWDS)

**Theorem:**

## What About IDWDS vs. NE and SNE?

Now try a modified matching pennies (attacker-defender) game:

|   | H    | T      | D     |
|---|------|--------|-------|
| H | 1, 0 | 0, 1   | 0, -1 |
| T | 0, 1 | 1, 0   | 0, 0  |
| D | 0, 0 | -1, -1 | 0, 0  |

$$IDSDS(G) = \{H, T, D\} \times \{H, T, D\} (=S)$$

$$NE(G) =$$

$$IDWDS(G) =$$

$$SNE(G) =$$

**Theorem:**  $SNE(G) \subseteq IDWDS(G)$  (Strict Nash always survives IDWDS)

**Theorem:**

## What About IDWDS vs. NE and SNE?

Now try a modified matching pennies (attacker-defender) game:

|   | H    | T      | D     |
|---|------|--------|-------|
| H | 1, 0 | 0, 1   | 0, -1 |
| T | 0, 1 | 1, 0   | 0, 0  |
| D | 0, 0 | -1, -1 | 0, 0  |

$$IDSDS(G) = \{H, T, D\} \times \{H, T, D\} (=S)$$

$$NE(G) =$$

$$IDWDS(G) = \{H, T\} \times \{H, T\}$$

$$SNE(G) =$$

**Theorem:**  $SNE(G) \subseteq IDWDS(G)$  (Strict Nash always survives IDWDS)

**Theorem:**

## What About IDWDS vs. NE and SNE?

Now try a modified matching pennies (attacker-defender) game:

|   | H    | T      | D     |
|---|------|--------|-------|
| H | 1, 0 | 0, 1   | 0, -1 |
| T | 0, 1 | 1, 0   | 0, 0  |
| D | 0, 0 | -1, -1 | 0, 0  |

$$IDSDS(G) = \{H, T, D\} \times \{H, T, D\} (=S)$$

$$NE(G) = \{(D, D)\}$$

$$IDWDS(G) = \{H, T\} \times \{H, T\}$$

$$SNE(G) =$$

**Theorem:**  $SNE(G) \subseteq IDWDS(G)$  (Strict Nash always survives IDWDS)

**Theorem:**

## What About IDWDS vs. NE and SNE?

Now try a modified matching pennies (attacker-defender) game:

|   | H    | T      | D     |
|---|------|--------|-------|
| H | 1, 0 | 0, 1   | 0, -1 |
| T | 0, 1 | 1, 0   | 0, 0  |
| D | 0, 0 | -1, -1 | 0, 0  |

$$IDSDS(G) = \{H, T, D\} \times \{H, T, D\} (=S)$$

$$NE(G) = \{(D, D)\}$$

$$IDWDS(G) = \{H, T\} \times \{H, T\}$$

$$SNE(G) = \emptyset$$

**Theorem:**  $SNE(G) \subseteq IDWDS(G)$  (Strict Nash always survives IDWDS)

**Theorem:**  $NE(G)$  and  $IDWDS(G)$  are not necessarily related by inclusion.

Can have  $NE(G) \subseteq IDWDS(G)$  or  $IDWDS(G) \subseteq NE(G)$  or  $IDWDS(G) \cap NE(G) = \emptyset$

## Putting It All Together

- $SNE(G) \subseteq NE(G) \subseteq IDSDS(G)$
- $SNE(G) \subseteq IDWDS(G) \subseteq IDSDS(G)$
- $NE(G)$  and  $IDWDS(G)$  can go either way

Which can be empty??

## Putting It All Together

- $SNE(G) \subseteq NE(G) \subseteq IDSDS(G)$
- $SNE(G) \subseteq IDWDS(G) \subseteq IDSDS(G)$
- $NE(G)$  and  $IDWDS(G)$  can go either way

Which can be empty??

- $SNE(G)$  can be empty

## Putting It All Together

- $SNE(G) \subseteq NE(G) \subseteq IDSDS(G)$
- $SNE(G) \subseteq IDWDS(G) \subseteq IDSDS(G)$
- $NE(G)$  and  $IDWDS(G)$  can go either way

Which can be empty??

- $SNE(G)$  can be empty
- $NE(G)$  can be empty

## Putting It All Together

- $SNE(G) \subseteq NE(G) \subseteq IDSDS(G)$
- $SNE(G) \subseteq IDWDS(G) \subseteq IDSDS(G)$
- $NE(G)$  and  $IDWDS(G)$  can go either way

Which can be empty??

- $SNE(G)$  can be empty
- $NE(G)$  can be empty
- $IDWDS(G)$  and  $IDSDS(G)$  will **never** be empty (by design)

## A Bigger Example

Suppose OSU donor Ratmir Timashev makes the following offer to 50 students. Each secretly writes down a request from  $\{\$10, \$20, \$30, \dots, \$100\}$

- If **5 or fewer** students ask for \$100, everyone gets what they asked for
- Otherwise, everyone gets nothing

All students are selfish and greedy. What are the Nash equilibria?

## A Bigger Example

There are two families of equilibria.

**(a) Seven or more students request \$100.**

Everyone gets nothing. Can anyone gain by switching alone?  
Is this a **strict** Nash equilibrium?

**(b) Exactly five request \$100, and all 45 others request \$90.**

Everyone is paid. Can anyone gain by switching alone?  
Is this a **strict** Nash equilibrium?

Question: Why is *exactly six* requesting \$100 not an equilibrium?

# Infinite Strategy Sets

# Cournot Competition

Shell and Speedway sell an identical good: a gallon of regular. Station  $i$  chooses quantity  $q_i \geq 0$  at constant unit cost  $c$ . The price is a *function* of total output:

$$P(q_1, q_2) = a - b(q_1 + q_2), \quad a > c, \quad b > 0$$

Shell's profit:

$$\pi_1(q_1, q_2) = P(q_1, q_2) q_1 - cq_1 = (a - c)q_1 - bq_1^2 - bq_1q_2$$

Strategy sets are  $[0, \infty)$  (infinite). But a Nash equilibrium is still just a pair  $(\bar{q}_1, \bar{q}_2)$  where neither station wants to move.

Cournot wrote this down in 1838, inventing Nash equilibrium about 120 years early, for this one class of games.

# Variables and Parameters

Not every letter in a model plays the same role.

**Parameters:**  $a, b, c$

- fixed numbers, known to everyone, chosen by nobody
- placeholders for a number that is already settled before anyone moves
- left as letters so one solution covers every version of the game

**Choice variables:**  $q_1, q_2$

- placeholders for a number that someone gets to pick
- these are the strategies:  $S_i = [0, \infty)$

**Derived values** (functions of the variables):  $P(q_1, q_2), \pi_i(q_1, q_2)$

- not fixed in advance, and nobody chooses them directly
- they are *functions*: feed in the choices, get a number out

For any letter, ask: could a player have made it come out differently?

## Maximizing, With Two Variables

From the math review: to maximize  $f(x)$ , set  $f'(x) = 0$ .

Shell's profit depends on *two* quantities:

$$\pi_1(q_1, q_2) = (a - c)q_1 - bq_1^2 - bq_1q_2$$

But Shell only chooses  $q_1$ . Speedway's  $q_2$  is not Shell's to pick.

So treat  $q_2$  as a fixed number and maximize over  $q_1$  alone.

# Partial Derivatives

A **partial derivative** is an ordinary derivative taken while every other variable is held constant, treated as though it were just a number.

$\frac{\partial f}{\partial x}$  : differentiate in  $x$ , treat  $y$  as a constant

The curly  $\partial$  is the only new thing. The rules are the ones you already know.

$$f(x, y) = 3x^2 + 5xy + y^3$$

$$\frac{\partial f}{\partial x} = \underline{\hspace{4cm}}$$

$$\frac{\partial f}{\partial y} = \underline{\hspace{4cm}}$$

## Why That Works

Same  $f(x, y) = 3x^2 + 5xy + y^3$ . Suppose  $y$  is stuck at 2:

$$f(x, 2) = 3x^2 + 10x + 8$$

That is a function of  $x$  alone. Differentiate it the ordinary way:

$$\frac{d}{dx}f(x, 2) = \underline{\hspace{10cm}}$$

Now compare it to  $\frac{\partial f}{\partial x}$  with  $y = 2$  substituted in.

Holding a variable constant and plugging in a constant are the same operation.

# Finding the Maximizer

Two conditions, in words first:

- **First order:** at the top, the slope in the x direction is flat
- **Second order:** the curve bends downward there, so flat means a peak rather than a valley

$$\underbrace{\frac{\partial f}{\partial x} = 0}_{\text{slope is flat}} \quad \text{and} \quad \underbrace{\frac{\partial^2 f}{\partial x^2} < 0}_{\text{bending downward}}$$

Skip the second condition and a flat point could just as easily be a minimum.

## Shell's Problem

$$\pi_1(q_1, q_2) = (a - c)q_1 - bq_1^2 - bq_1q_2$$

Hold  $q_2$  fixed. First order:

$$\frac{\partial \pi_1}{\partial q_1} = \underline{\hspace{2cm}} = 0$$

Second order:

$$\frac{\partial^2 \pi_1}{\partial q_1^2} = \underline{\hspace{2cm}}, \text{ negative because } b > 0, \text{ so the flat point is a maximum.}$$

## From First-Order Condition to Best Reply

Shell's first-order condition holds at Shell's optimum, *whatever*  $q_2$  turns out to be:

$$a - c - 2bq_1 - bq_2 = 0$$

So solve it for  $q_1$ :

$$BR_1(q_2) = \frac{a - c - bq_2}{2b}$$

The second-order condition was  $-2b < 0$ , so this really is Shell's best, not its worst.

## A Best Reply Is a Function

$$BR_1(q_2) = \frac{a - c - bq_2}{2b}$$

Feed it a quantity for Speedway, and it hands back Shell's best answer to that quantity. One input, one output.

- If Speedway floods the market,  $BR_1$  tells Shell to pull back
- If Speedway sits out,  $BR_1$  tells Shell to act like a monopolist

Speedway's problem is the same with the labels swapped:

$$BR_2(q_1) = \frac{a - c - bq_1}{2b}$$

$BR_i(q_{-i})$  is truncated at 0 if needed.

## Nash Equilibrium Is a Fixed Point

Recall the best-reply version of the definition:  $(q_1^*, q_2^*)$  is a Nash equilibrium exactly when each is a best reply to the other.

$$q_1^* = BR_1(q_2^*) \quad \text{and} \quad q_2^* = BR_2(q_1^*)$$

Substitute the second into the first:

$$q_1^* = BR_1(BR_2(q_1^*))$$

Two equations in two unknowns just became **one** equation in one unknown.

Start at  $q_1^*$ , ask what Speedway would do, then ask what Shell would do back. A Nash equilibrium is a quantity that returns to *itself* after that round trip.

## Solving the Fixed Point

$$q_1 = \frac{a - c - b BR_2(q_1)}{2b}, \quad BR_2(q_1) = \frac{a - c - bq_1}{2b}$$

Substitute, then solve for  $q_1$ :

## And Then the Other One

Feed  $q_1^*$  back through Speedway's best reply:

$$q_2^* = BR_2(q_1^*) = \frac{a - c - b \left( \frac{a-c}{3b} \right)}{2b}$$

$$q_2^* = \underline{\hspace{2cm}}$$

No surprise that it matches  $q_1^*$ : the two stations face identical problems, so the equilibrium had to be symmetric.

Worth checking directly that  $BR_1(q_2^*) = q_1^*$ . It does.

## Cournot: The Numbers

At  $\bar{q}_1 = \bar{q}_2 = \frac{a - c}{3b}$ :

- Price:  $P(\bar{q}_1, \bar{q}_2) = \frac{a + 2c}{3}$

- Each profit:  $\frac{(a - c)^2}{9b}$

Try it:  $a = 25$ ,  $b = 2$ ,  $c = 1$ .

$$\bar{q}_1 = \bar{q}_2 = \underline{\hspace{2cm}}$$

$$P = \underline{\hspace{2cm}}$$

$$\text{profit} = \underline{\hspace{2cm}}$$

**Let's Work One Out**

## Three Bars on High Street

Ann, Ben, and Cal each run a bar. Before a home game each independently commits to a **Big** promotion or a **Small** one.

|     |       | Ben      |         |
|-----|-------|----------|---------|
|     |       | Big      | Small   |
| Ann | Big   | 2, 2, 2  | 0, 3, 1 |
|     | Small | 3, 0, 1  | 1, 1, 0 |
|     |       | Cal: Big |         |

|     |       | Ben        |         |
|-----|-------|------------|---------|
|     |       | Big        | Small   |
| Ann | Big   | 1, 1, 3    | 0, 0, 2 |
|     | Small | 0, 2, 0    | 2, 2, 3 |
|     |       | Cal: Small |         |

Ann picks the row, Ben the column, Cal the table. Payoffs in order: Ann, Ben, Cal.

## Three Bars on High Street

1. Does anyone have a dominant strategy?
2. Does IDSDS delete anything at all?
3. Find every Nash equilibrium. Underline Ann's best payoff in each column, Ben's in each row, and Cal's whenever it beats her payoff in the same cell of the other table.
4. Is either equilibrium Pareto superior to the other?

## What Have We Learned, Charlie Brown?

- **Second-price auction:** truthful bidding is weakly dominant (Vickrey)
- **Pivotal mechanism:** truthful reporting is weakly dominant (Clarke)
- **IDSDS:** delete strictly dominated strategies, repeat (order-free, justified by common knowledge of rationality)
- **IDWDS:** order matters, so delete simultaneously, and the justification is murkier
- **Nash equilibrium:** nobody gains by deviating alone
- $NE \subseteq IDSDS$ , but a game may have no Nash equilibrium at all

Next: what happens when players can randomize.

## Reading & Practice

- Bonanno, *Game Theory*, §2.3–2.7
- Exercises: §2.9.3 (auctions), §2.9.4 (pivotal mechanism), §2.9.5 (iterated deletion), §2.9.6 (Nash equilibrium), §2.9.7 (infinite games)
- Proofs of both theorems are in §2.8
- Solutions in §2.10. Try them first